ST 3007 – Operational Research

Tutorial 1

Integer Programming

1. Solve the following IP problems **Gomory's Cutting Plane Algorithm.**

   a) Maximize $z = 3x_1 + 4x_2$
   Subject to
   $2/5x_1 + x_2 \leq 3$
   $2/5x_1 + 2/5x_2 \leq 1$
   $x_1, x_2 \geq 0$ and integers

   b) Maximize $z = 2x_1 + 3x_2$
   Subject to
   $x_1 + 2x_x \leq 3$
   $4x_1 + 5x_2 \leq 10$
   $x_1, x_2 \geq 0$ and integers

2. Solve the following IP problems using **B&B method** and draw the **B&B tree.**

   a) Maximize $z = 3x_1 + 2x_2$
   Subject to
   $2x_1 + 2x_2 \leq 9$
   $3x_1 + 3x_2 \leq 18$
   $x_1, x_2 \geq 0$ and integers

   b) Maximize $z = 5x_1 + 4x_2$
   Subject to
   $x_1 + x_2 \leq 5$
   $10x_1 + 6x_2 \leq 45$
   $x_1, x_2 \geq 0$ and integers